Planning Report (Draft)

An Open Project in Deep Learning: Hand Gesture Recognition System

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1 Objective

The project goal is to develop a hand gesture recognition system.

Secondary goals if primary goal is achieved:

- Enable custom gesture addition without retraining the network.
- Match and annotate gestures from videos.

2 Rationale

While most common gesture recognition models focuses on a fixed set of gestures [1], this project aims to improve the flexibility and usability of gesture recognition systems, with the following sub-goals:

- Address the need of custom gestures.
- Automate the annotation process for unannotated videos and allow developers to efficiently annotate certain gestures into the database.

3 Approach and Timeline

The project will be executed in several phases, with increasing complexities and depth.

3.1 Phase I

The first phase involves the following:

- Review traditional gesture recognition techniques, where:
  - The model is trained to recognize a fixed set of gestures, and
  - A single gesture corresponds to one action/one meaning.
- Utilize a basic model for gesture recognition.
- Define input/output formats.
- Use DWPose[2]/MediaPipe[3]/HRNet[4] for keypoint conversion with pretrained models and use keypoints for gesture recogntion.

In this phase, we conduct basic research in the field and verify that the basic framework is working.

3.2 Phase II

The second phase involves the following:

- Develop upon the previous phase, train a more complex model with backbones preferrably pre-trained.
- Investigate eye-tracking integration.
- Handle gestures involving both hands, where each hand can correspond to a separate action.
- Conduct literature research on the following:
  - Enhance prediction accuracy with additional models and knowledge distillation.
  - Refer to relevant papers like TwoStreamNetwork (SLRT)[5] and SlowFastSign[6] for sign language, and YOLO-v8[7][8] for gestures.

In this phase, we delve into more comprehensive literature research and started to develop our own models.

3.3 Phase III

The third phase corresponds to the first of the secondary goals, which involves the following:

- Allow users to add custom gestures.
- Avoid the need for network retraining for new gestures.
- Research and review for feasibility.

In this phase, we would start focusing on transferring our model such that it can be utilized for various gesture-related tasks.

3.3.1 Previous Work

Pretrained framework for fine-tuning on customized set of gestures. While most existing hand gesture recognition (HGR) systems are limited to a predefined set of gestures, Uboweja et al. presented a framework that allowed users to fine-tune the model on custom set of gestures with low data and resource requirements [1].

3.4 Phase IV

The fourth phase corresponds to the second of the secondary goals, which involves the following:

- Given annotated video, match and append new gesture-gloss pairs to a database.
- For unannotated videos, match and annotate with glosses, and handle unknown postures.
- Ultimately performs an easy pipeline for adapting gesture recognition on sign language translation tasks, even on datasets like the Hong Kong Sign Language (HKSL) with fairly few samples to generalize on.

While we may not have sufficient time to finish the project up to this phase, ultimately we expect to produce a high-performance model for complex gesture recognition tasks like sign language translation with low resource requirements for the users.

4 Possible Issues

4.1 Possible obstacles

We expected the following obstacles in the process of our work:

- **Data limitations.** Insufficient data for different viewing angles.
- **Complexity.** The project may be too complicated for a small team.
- **Composite signs.** Handling signs with composite meanings or specific facial expressions.

- **Left-right hand differentiation.** Managing messages conveyed by different hands simultaneously.
- **Blurriness.** Addressing issues with blurry input data.

For phase IV, we further expect to encounter the following obstacles:

- **Data limitations.** Insufficient data for the HKSL dataset. Most sources are not open for public use. We expect to utilize a very small subset of the German Sign-Language Weather Phoenix-2014t dataset to evaluate the performance of our model if we failed to gather sufficient training data on the HKSL dataset.

4.2 Backup plans

In case that we failed to achieve a satisfactory accuracy with our models, the project could still be finished with a few achievements:

- Develop a useful product with a user interface for camera capture.
- Review and evaluate previous models academically.
- Test and evaluate the scores and performances on existing models.

References

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